Manju Swaroop Venkataravanappa

manjuswaroop03@gmail.com
LinkedIn | GitHub

EDUCATION

Queen Mary University of London *2026 – 2027*

MSc Data Science
Modules: Machine Learning, NLP, Data Mining, Big Data Processing, Neural Networks and Deep Learning, Applied Statistics

Dayananda Sagar University, Bengaluru, India ***8.82 CGPA*** | *2020 – 2024*

Bachelor of Technology, CSE, Specialisation in Data Science

WORK EXPERIENCE

Associate Consultant • Ernst & Young LLP *Jul 2024 – Sep 2026*

- Validated and optimised complex **ETL data pipelines** across 50+ interfaces using **SQL, Python**, and **Apache Spark on Databricks** to transform transactional data, achieving 98% data accuracy through validation checks across FMCG, Retail, and Oil & Gas sectors.
- Developed **AI-driven classification models** and automated **data workflows**, using **Kafka** to stream data between systems, to streamline compliance processes (SPL Screening, Legal Control) for a greenfield landscape.
- Processed, cleaned, and validated 120+ country-specific classification datasets from multiple source systems, storing raw and processed data in **Amazon S3** and adding **data quality checks** to support model evaluation and audit reviews.
- Prepared features, trained and tested **machine learning models**, and improved them using evaluation results; explained findings to team leads and clients in plain language.

Data Analyst Intern • Village Company *Jan 2024 – Jun 2024*

- Analysed customer retention and **churn rates** for 500+ premium clients using **SQL** to identify key post-revenue growth drivers.
- Developed **AI predictive models** and cohort analyses in Python to optimise customer retention strategies for luxury goods.
- Wrote **SQL queries** to extract and clean customer data, and shared findings with the team to support business decisions.

Data Scientist Intern • Amazon *Jul 2022 – Sep 2022*

- Devised and implemented **data analysis and statistical techniques** to develop solutions and data-driven insights.
- Cleaned and prepared user and location data in **Python** for modelling and analysis.
- Worked on a **Dynamic Pricing problem** to predict the suitable price using previous data history of user and location, achieving a 10% increase in average transaction value and improving service provider retention by 25%.
- Used **web scraping techniques** to extract and organise competitor data, and built data visualisations to provide actionable insights.

TECHNICAL SKILLS

Languages	Python, SQL, R
Data Engineering	Apache Spark, Kafka, Databricks, Amazon S3
Tools	Git, Tableau, Power BI, RapidMiner, Microsoft Excel, Altair
Libraries	Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, Scrapy

PROJECTS

Sensitive Data Classification using Deep Network with Sequential Models

- Implemented and fine-tuned deep learning models such as **Convolutional Neural Network (CNN)** and **Long Short-Term Memory (LSTM)** to **classify sensitive data** with high accuracy.
- Compared model performance of **CNN, SVM,** and **Logistic Regression**, with the CNN outperforming traditional methods by over 13.5% in accuracy (**CNN: 97.5%, SVM: 84.02%**). Trained over 500 epochs using **binary cross-entropy loss** and **Adam optimiser**, resulting in a smooth convergence curve and minimised overfitting risk.

Gene Expression Analysis for Leukemia

- Developed a machine learning pipeline for early **leukemia detection** by leveraging gene expression data, integrated **Rat Swarm Optimisation (RSO)** for efficient feature selection, which improved classification **accuracy to 98.12%**.
- Implemented and compared 3 ML models (Ridge, Passive Aggressive, Stochastic Gradient), where Ridge with RSO showed **96.85% precision** and **94.84% Matthews correlation coefficient**.

Personalised AI Assistant with Contextual Recall

- Developed a custom **conversational AI agent** to provide personalised, coherent responses using **LangChain** and **LangGraph**, and integrated **vector-based memory** (**Chroma DB**) with contextual memory, improving response continuity by **over 85%**.
- Deployed the assistant in a local runtime using **FastAPI**, with real-time interaction logs and memory visualisations, achieving **sub-300ms response times** on average over 100+ test cases.

CERTIFICATIONS

- **Databricks Certified Data Engineer Associate** – covers Apache Spark (PySpark, Spark SQL), Delta Lake, and ETL pipelines on the Databricks Lakehouse.
- EY **Agentic AI** – Bronze Learning (2026), offered by "**Ernst & Young LLP**".
- NPTEL certification "**Data Science for Engineers**", offered by "**IIT Madras**". Got Elite Silver rating in this course (Apr 2023).
- Coursera certification "**Machine Learning Specialization**", offered by "**Stanford University**".
- EY **Artificial Intelligence Applied AI** – Bronze Learning (2024), offered by "**Ernst & Young LLP**".
- Coursera certification on "**Introduction to Data Science in Python**", offered by "**University of Michigan**".

ACHIEVEMENTS

- Presented research paper on "**Gene Expression Analysis for Leukemia**" at **ICCCNT 2023**, IIT Delhi.
- Published paper on "**Sensitive Data Classification using Deep Network with Sequential Models**" at **ICRTC 2023**, Ghaziabad.
- Recognised as a **NASA International Space Apps Global Nominee** in 2022, for solutions to Earth-related challenges.
- **1st Position** in "Altair Expo" conducted by Dayananda Sagar University, Bangalore. Implemented **predictive modelling algorithms** in RapidMiner to analyse customer behaviour and identify factors contributing to churn.
- Neo Coding Arena **Challenge Winner**.